

AI Audit Report

Mandatory appendix for AI-assisted coursework.

This Markdown format is adapted from the FIT@HCMUS AI Audit Report - 5-section Template per Artifact.

1. Student Information

Field	Value
Student name (printed)	
Student ID	
Class / Cohort	
Assignment ID	
Assignment date	
AI tool(s) used	
AI assistance declared	☐ Yes ☐ No

Student Information is intentionally left blank for student completion.

2. Instructions

- Add one audit entry for each AI-generated artifact or meaningful prompt-output interaction.
- Preserve verbatim prompts and AI outputs in docs/ai-audit/interactions/ when long.
- Assign VALID, INVALID, or INCOMPLETE only after human review.
- Record Student Fix and verification separately from the AI verdict.
- Do not remove or rewrite historical entries.

3. Audit Entries - Five Sections per Artifact

```
<!-- AUDIT ENTRIES START -->
```

Artifact A-001 - FR-05 clean-restart requirement analysis and test-case design

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-09 23:26:56 +07:00
Workflow Stage	Requirement analysis and test-case design
Feature / Task	FR-05 - Product listing and search
Related Artifact	docs/test-cases/fr-05/

Verbatim Prompt

[interactions/A-001-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-001-output.md

Verbatim AI Output or Labelled Excerpt

Status: TEST_CASE_DESIGN_REVIEW_REQUIRED
 Test Cases: 16
 Automation Candidate Count: 12

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	INCOMPLETE
Verdict Scope	FR-05 requirement analysis and test-case design generated before human corrections

(4) Reasoning

Review Notes

- Requirement analysis bám đúng các authoritative/supporting sources và không dùng implementation để thay đổi Expected Result.
- Test design đã loại bỏ arbitrary boundary data và ưu tiên existing verified seed data.
- Các behavior có automation feasibility khác nhau nhìn chung đã được tách hợp lý.
- FR05-TC-001 vẫn kết hợp catalog completeness và grid layout dù hai behavior có thể fail độc lập; cần split theo Atomic Test Objective Rule.
- FR05-TC-006 nên đổi từ EDGE sang POSITIVE vì case hiện dùng normal existing seed prices và không còn kiểm tra boundary.
- Sau các chỉnh sửa trên, automation candidate count dự kiến tăng từ 12 lên ít nhất 13, tạo buffer tốt hơn trước automation planning.

(5) Student Fix

Field	Value
Student Decision	MODIFIED
Change Illustration	See Changes Made below.
Verification Method	REVIEW
Verification Result	PASSED
Final File	docs/test-cases/fr-05/
Approval Status	APPROVED

Changes Made

- Split catalog completeness/count from grid-layout verification.
- Added FR05-TC-017 for the independent grid-layout objective.
- Reclassified FR05-TC-006 from EDGE to POSITIVE.
- Recalculated test-case and automation-feasibility statistics.
- Final corrected design contains 18 test conditions, 17 test cases, and 13 automation candidates.

Correction Notes

The original AI output required human-directed corrections, therefore the original verdict remains INCOMPLETE. The corrected artifact was reviewed again and approved for downstream automation planning.

Human Decision Evidence

APPROVE TEST CASE DESIGN

Artifact A-002 - FR-05 Playwright automation plan

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	gpt-5.6-sol
Date and Time	2026-08-09T17:20:32.949Z
Workflow Stage	Automation planning; before human approval
Feature / Task	FR-05 - Product listing and search
Related Artifact	docs/automation-plans/fr-05-automation-plan.md

Verbatim Prompt

[interactions/A-002-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-002-output.md

Verbatim AI Output or Labelled Excerpt

1. **Approved Test Cases Received:** **17**
2. **Automation Minimum Pre-check Result:** **PASS**
3. **Automation Candidate Count:** **13**, đạt minimum **12**
21. **Current Checkpoint:** **CHECKPOINT: AUTOMATION_PLAN_REVIEW_REQUIRED**

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	INCOMPLETE
Verdict Scope	FR-05 Playwright automation plan before human-directed corrections

(4) Reasoning

Review Notes

The original AI-generated automation plan required human-directed corrections, therefore the original verdict remains INCOMPLETE.

After correction and human review, the final plan contains:

- 17 approved test cases
- 13 automation candidates
- 2 READY_FOR_AUTOMATION
- 11 READY_WITH_SETUP
- 4 BLOCKED_BY_IMPLEMENTATION

- Automation Minimum Pre-check: PASS

(5) Student Fix

Field	Value
Student Decision	MODIFIED
Change Illustration	Original plan required human correction for TC-006 grouping and TC-008 result oracle.
Correction Notes	See Changes Made below.
Verification Method	REVIEW
Verification Result	PASSED
Final File	docs/automation-plans/fr-05-automation-plan.md
Approval Status	APPROVED

Changes Made

- Strengthened FR05-TC-008 so the primary result oracle is product result COUNT = 0.
- Kept description-only search behavior independent from empty-state presentation.
- Strengthened FR05-TC-006 thousand-grouping validation.
- Restricted accepted thousand grouping separators to conventional grouping characters.
- Required separator consistency and rejected raw digits, letters, hyphens, and mixed separators.
- Revalidated all 17 approved test cases.
- Confirmed 13 legitimate automation candidates remain.

Human Decision Evidence

APPROVE AUTOMATION PLAN

Artifact A-003 - FR-05 external test-data preparation

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	gpt-5.6-sol
Date and Time	2026-08-09T17:54:52.588Z
Workflow Stage	Test Data Design / Preparation; before human approval
Feature / Task	FR-05 - Product listing and search
Related Artifact	test-data/fr-05.json; docs/test-data/fr-05-test-data-plan.md

Verbatim Prompt

[interactions/A-003-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-003-output.md

Verbatim AI Output or Labelled Excerpt

1. **Test Data File:** [fr-05.json](<D:/Workspace/HCMUS/Software Testing/Homework/group05_eshop/test-data/fr-05.json>)
2. **Test Data Plan File:** [fr-05-test-data-plan.md](<D:/Workspace/HCMUS/Software Testing/Homework/group05_eshop/docs/test-data/fr-05-test-data-plan.md>)
3. **Total Datasets:** 8 logical datasets - 7 data datasets và 1 environment setup dataset.

13. ****Current Checkpoint:**** `CHECKPOINT: TEST\_DATA\_REVIEW\_REQUIRED`

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	INCOMPLETE
Verdict Scope	FR-05 external test-data preparation before human-directed correction

(4) Reasoning

Review Notes

The original AI output required one human-directed test-setup correction, therefore its verdict remains INCOMPLETE.

The corrected data package contains:

- 9 logical datasets
- 7 data datasets
- 2 setup datasets
- 13/13 automation candidates covered
- 5 verified seed products reused
- 0 new database records
- 4 BLOCKED_BY_IMPLEMENTATION cases preserved

(5) Student Fix

Field	Value
Student Decision	MODIFIED
Change Illustration	Original output contained safe-rendering input strings but not the shared controlled empty-response setup.
Changes Made	See Changes Made below.
Correction Notes	The original AI output required one human-directed test-setup correction, therefore its verdict remains INCOMPLETE.
Verification Method	REVIEW
Verification Result	PASSED
Final File	test-data/fr-05.json; docs/test-data/fr-05-test-data-plan.md
Approval Status	APPROVED

Changes Made

- Added FR05-SETUP-002 as a reusable controlled empty product-search response setup.
- Mapped the setup to FR05-TC-010, FR05-TC-011 and FR05-TC-012.
- Externalized GET /api/products responseBody [] instead of leaving the future response fixture inline.
- Preserved host/base URL as runtime configuration.
- Revalidated all 13 automation candidates.
- Preserved all four BLOCKED_BY_IMPLEMENTATION cases.

Human Decision Evidence

APPROVE TEST DATA

Artifact A-004 - FR-05 Playwright script generation and AI-generated code review

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	gpt-5.6-sol
Date and Time	2026-08-09T18:24:06.1289590Z
Workflow Stage	Playwright script generation and AI-generated code review; before human automation review
Feature / Task	FR-05 - Product listing and search
Related Artifact	tests/fr-05/; playwright.config.js; docs/automation-reviews/fr-05-ai-review.md; docs/gaps/fr-05-automation-gaps.md

Verbatim Prompt

[interactions/A-004-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-004-output.md

Verbatim AI Output or Labelled Excerpt

1. ****Test Scripts Created:**** [fr-05.spec.js](<D:/Workspace/HCMUS/Software Testing/Homework/group05_eshop/tests/fr-05/fr-05.spec.js>)
2. ****Automation Script Count:**** 13 Playwright tests.
18. ****Current Checkpoint:**** `CHECKPOINT: AUTOMATION\_REVIEW\_REQUIRED`

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	INCOMPLETE
Verdict Scope	FR-05 Playwright script generation and AI-generated code review before human automation review

(4) Reasoning

Review Notes

The original FR-05 Playwright script-generation and AI code-review output required human-directed corrections, therefore its verdict remains INCOMPLETE.

The corrected automation artifact was statically reviewed and approved. The corrections addressed data traceability, visible HTML reporter identity, accepted locator-risk classification, and execution-readiness boundaries. Runtime Playwright execution remains unverified, and FR05-REV-006 remains NEEDS_MORE_EVIDENCE until multi-browser execution.

(5) Student Fix

Field	Value
Student Decision	MODIFIED
Change Illustration	Human review corrected TC-002 dataset traceability and strengthened visible HTML report identity while preserving the approved automation scope.
Changes Made	See Changes Made below.
Correction Notes	The original AI output required human-directed corrections, therefore its verdict remains INCOMPLETE.
Verification Method	STATIC_REVIEW
Verification Result	PASSED
Final File	tests/fr-05/; playwright.config.js; docs/automation-reviews/fr-05-ai-review.md; docs/gaps/fr-05-automation-gaps.md
Approval Status	APPROVED

Changes Made

- Preserved explicit controlled-route cleanup introduced during generation.
- Corrected FR05-TC-002 data traceability so it consumes FR05-DATA-001.
- Preserved FR05-DATA-002 exclusively for FR05-TC-007.
- Added a runtime-derived visible HTML reporter title containing Feature ID, Run by: <StudentID>, ISO timestamp, and Run ID.
- Preserved the existing reporter metadata.
- Classified locator risks as accepted documented risks.
- Deferred Playwright dependency/browser installation to Execution Readiness.
- Preserved multi-browser dialog timing as NEEDS_MORE_EVIDENCE.

Human Decision Evidence

APPROVE AUTOMATION REVIEW FIXES

Artifact A-005 - FR-05 Execution Readiness**(1) Prompt + Tool**

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T01:57:18.0306121+07:00
Workflow Stage	Execution readiness; before multi-browser execution
Feature / Task	FR-05 - Product listing and search
Related Artifact	docs/execution-readiness/fr-05-execution-readiness.md; package.json; package-lock.json

Verbatim Prompt[interactions/A-005-prompt.md](#)**(2) AI Output**

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-005-output.md

Verbatim AI Output or Labelled Excerpt

1. **Execution Readiness Result:** `READY\_FOR\_EXECUTION`

13. **Test Discovery Result:** `PASS` - `--list` exited `0`, collected one spec and 39 project-test combinations. No tests were executed.

21. **Current Checkpoint:** `CHECKPOINT: EXECUTION\_READINESS\_REVIEW\_REQUIRED`

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-05 Execution Readiness preparation

(4) Reasoning

Review Notes

Human review confirmed that the execution-readiness preparation correctly installed and resolved Playwright, verified all three browser binaries, accepted the Student ID and SUT base URL, discovered the approved tests under each browser project, preserved retries and reporter identity, and did not execute runtime tests. Runtime SUT availability and seed-state verification remain execution preconditions.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Change Illustration	No human-directed correction was required.
Changes Made	No human-directed correction was required.
Correction Notes	Human review accepted the readiness preparation as produced.
Verification Method	REVIEW
Verification Result	PASSED
Final File	docs/execution-readiness/fr-05-execution-readiness.md; package.json; package-lock.json
Approval Status	APPROVED

Verification Notes

- @playwright/test 1.62.1 installed.
- chromium, firefox and webkit installed.
- STUDENT_ID=23127107 accepted.
- SUT_BASE_URL=http://localhost:5173 resolved.
- 13 approved FR-05 Test Case IDs discovered.
- 13 tests discovered under each browser project.
- 39 total project-test combinations.
- Four implementation-blocked IDs absent.
- retries remains 0.
- HTML reporter identity strategy retained.
- No runtime test execution occurred during readiness.

Execution remains subject to runtime SUT availability and seed-state verification immediately before the approved browser run.

Human Decision Evidence

APPROVE EXECUTION READINESS

Artifact A-006 - FR-05 Chromium execution and initial failure triage

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-09T19:20:49.2608303Z
Workflow Stage	Chromium runtime execution before human execution review
Feature / Task	FR-05 - Product listing and search
Related Artifact	docs/execution-results/fr-05-chromium-execution.md; Chromium HTML report; test-results/fr-05/

Verbatim Prompt

[interactions/A-006-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-006-output.md

Verbatim AI Output or Labelled Excerpt

1. **Chromium Execution Status:** `COMPLETED\_WITH\_FAILURES` - 13 tests completed in 13.5s; Playwright exit code `1` reflects test a
14. **Report Identity Verification:** `PASS` - the real rendered report title/body contains `FR-05`, `Run by: 23127107`, `2026-08-0
21. **Current Checkpoint:** `CHECKPOINT: CHROMIUM\_EXECUTION\_REVIEW\_REQUIRED`

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	Original FR-05 Chromium execution and initial failure triage

(4) Reasoning

Review Notes

Human review confirmed that the original Chromium execution record accurately preserved all 13 approved test results, the six initial classifications, rendered report identity evidence, Chromium dialog/DOM evidence for TC-012, and the documented backend-startup database side effect. No environment failure occurred. The TC-006 automation correction remains a separate follow-up before later cross-browser execution.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Change Illustration	No correction to the original Chromium execution record was required.
Changes Made	No correction to the original Chromium execution record was required.
Correction Notes	Human review accepted the original Chromium execution and initial triage.
Verification Method	REVIEW
Verification Result	PASSED
Final File	docs/execution-results/fr-05-chromium-execution.md
Approval Status	APPROVED

Verification Notes

- Chromium execution completed all 13 approved automated tests.
- 7 tests passed and 6 tests failed.
- FR05-TC-004 is PRODUCT_DEFECT.
- FR05-TC-005 is PRODUCT_DEFECT.
- FR05-TC-006 is AUTOMATION_DEFECT.
- FR05-TC-011 is PRODUCT_DEFECT.
- FR05-TC-012 is PRODUCT_DEFECT.
- FR05-TC-014 is PRODUCT_DEFECT.
- HTML report identity verification passed.
- FR05-TC-012 captured real Chromium dialog and executable DOM evidence.
- No environment failure occurred.
- backend/database.sqlite modification is a documented normal SUT startup side effect and is not classified as an automation defect.

APPROVE CHROMIUM EXECUTION TRIAGE

Artifact A-007 - FR-05 runtime automation correction for FR05-TC-006 and Chromium verification

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-09T19:45:07.9247710Z
Workflow Stage	Runtime automation correction after Chromium failure triage
Feature / Task	FR-05 - Product listing and search
Related Artifact	tests/fr-05/helpers/fr-05-helpers.js; docs/automation-reviews/fr-05-ai-review.md; docs/execution-results/fr-05-chromium-rerun-after-tc006-fix.md; Chromium HTML report

Verbatim Prompt

[interactions/A-007-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-007-output.md

Verbatim AI Output or Labelled Excerpt

11. FR05-TC-006 Result

PASSED - it no longer fails due to the grouping-helper logic.

18. Current Checkpoint

CHECKPOINT: TC006_CORRECTION_REVIEW_REQUIRED

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-05 runtime automation correction for FR05-TC-006 and Chromium verification

(4) Reasoning

Review Notes

Human review confirmed the TC-006 automation correction without changing the approved Expected Result or introducing a VND-specific special case. Five approved grouping formats passed static verification, five invalid/raw/mixed formats were rejected, runtime seed state matched the approved oracle, Chromium rerun completed all 13 tests with 8 passed and 5 failed, TC-006 changed from FAILED to PASSED, no unrelated regression occurred, the five remaining failures remained the previously confirmed PRODUCT_DEFECT cases, HTML report identity passed, and original Chromium evidence remained preserved. The original prompt and full AI output remain preserved externally.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Change Illustration	No additional human-directed correction was required after the A-007 output.
Changes Made	No additional human-directed correction was required after the A-007 output.
Correction Notes	Human review confirmed the TC-006 correction and Chromium verification as recorded in the final files.
Verification Method	STATIC_REVIEW_AND_RUNTIME_RERUN
Verification Result	PASSED
Final File	tests/fr-05/helpers/fr-05-helpers.js; docs/automation-reviews/fr-05-ai-review.md; docs/execution-results/fr-05-chromium-rerun-after-tc006-fix.md
Approval Status	APPROVED

Verification Notes

- FR05-TC-006 passed after the helper correction.
- Chromium rerun recorded 8 passed and 5 failed tests.
- The remaining five failures are the previously confirmed PRODUCT_DEFECT cases: FR05-TC-004, FR05-TC-005, FR05-TC-011, FR05-TC-012, and FR05-TC-014.
- HTML report identity verification passed.
- Firefox and WebKit were not run.

Human Decision Evidence

APPROVE TC006 AUTOMATION CORRECTION

Artifact A-008 - FR-05 Firefox/WebKit execution and cross-browser comparison

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-09T20:03:43.0434782Z
Workflow Stage	Remaining multi-browser execution before final defect reporting
Feature / Task	FR-05 - Product listing and search
Related Artifact	docs/execution-results/fr-05-firefox-execution.md; docs/execution-results/fr-05-webkit-execution.md; docs/execution-results/fr-05-cross-browser-summary.md; Firefox/WebKit HTML reports; docs/automation-reviews/fr-05-ai-review.md

Verbatim Prompt

[interactions/A-008-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-008-output.md

Verbatim AI Output or Labelled Excerpt

17. Chromium / Firefox / WebKit Comparison

Each engine completed 13 tests with 8 passed and 5 failed. FR05-TC-006 passed across all three.

24. Current Checkpoint

CHECKPOINT: CROSS_BROWSER_EXECUTION_REVIEW_REQUIRED

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-05 Firefox/WebKit execution and cross-browser comparison

(4) Reasoning

Review Notes

Human review accepted the cross-browser execution record. Firefox and WebKit each completed 13 tests with 8 passed and 5 failed; the five failures matched the approved Chromium baseline PRODUCT_DEFECT cases; TC-006 passed across all three engines; no new AUTOMATION_DEFECT or ENVIRONMENT_FAILURE was found; all three HTML reports passed identity verification; and FR05-REV-006 reached RUNTIME_VERIFIED_MULTI_BROWSER. The original prompt and full AI output remain preserved externally.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Change Illustration	No human-directed correction was required.
Changes Made	No human-directed correction was required.
Correction Notes	Human review confirmed the final cross-browser execution and comparison evidence, including the browser-specific TC-012 dialog count.
Verification Method	MULTI_BROWSER_RUNTIME_REVIEW
Verification Result	PASSED
Final File	docs/execution-results/fr-05-firefox-execution.md; docs/execution-results/fr-05-webkit-execution.md; docs/execution-results/fr-05-cross-browser-summary.md; docs/automation-reviews/fr-05-ai-review.md
Approval Status	APPROVED

Verification Notes

- Firefox: 13 total, 8 passed, 5 failed.
- WebKit: 13 total, 8 passed, 5 failed.
- FR05-TC-006 passed on Chromium, Firefox, and WebKit.
- The five remaining failures are the same confirmed PRODUCT_DEFECT cases on all engines.
- FR05-REV-006 is RUNTIME_VERIFIED_MULTI_BROWSER.
- HTML report identity verification passed independently for all three engines.

Human Decision Evidence

APPROVE CROSS BROWSER EXECUTION

Artifact A-009 - FR-05 final defect reporting and feature closure preparation**(1) Prompt + Tool**

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-09T20:23:46.0596808Z
Workflow Stage	Final FR-05 review before GitHub issue publication / feature closure
Feature / Task	FR-05 - Product listing and search
Related Artifact	docs/defects/fr-05/; docs/execution-results/fr-05-final-summary.md

Verbatim Prompt[interactions/A-009-prompt.md](#)**(2) AI Output**

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-009-output.md

Verbatim AI Output or Labelled Excerpt**1. Confirmed Defects**

Exactly five confirmed PRODUCT_DEFECT records were created: FR05-TC-004, FR05-TC-005, FR05-TC-011, FR05-TC-012, and FR05-TC-014. No

17. Current Checkpoint

CHECKPOINT: FR05_FINAL_REVIEW_REQUIRED

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	INCOMPLETE
Verdict Scope	Original FR-05 final defect reporting and feature-closure bundle before human-directed defect-format and language normalization.

(4) Reasoning

Review Notes

- Original A-009 xác định đúng năm confirmed PRODUCT_DEFECT cases, tạo defect reports, screenshots, GitHub Issue drafts và final FR-05 summary.
- Các defect report ban đầu dùng Test Case ID làm Defect ID, chưa tuân theo HW04 Bug Report Standard về structure, filenames, screenshots, GitHub Issue drafts, hashtags, labels và ngôn ngữ mô tả.
- Điều này làm bundle chưa sẵn sàng để handoff theo standard, nhưng không làm thay đổi execution evidence hoặc product-defect classification đã được xác nhận.
- AI có thể bỏ sót vì output ban đầu tập trung vào việc ghi nhận defect evidence và chưa áp dụng đầy đủ standard hoá Defect ID/bug-report format ở bước final handoff.
- Human-directed correction đã chuẩn hoá bundle mà không rerun browser/test, recapture screenshot hoặc thay đổi SUT.
- Nguồn đánh giá: human review và final bundle tại docs/defects/fr-05/ cùng docs/execution-results/fr-05-final-summary.md.

(5) Student Fix

Field	Value
Student Decision	MODIFIED
Change Illustration	See Change Illustration below.
Changes Made	See Changes Made below.
Correction Notes	See Correction Notes below.
Verification Method	REVIEW
Verification Result	PASSED
Final File	docs/defects/fr-05/; docs/execution-results/fr-05-final-summary.md
Approval Status	APPROVED

Changes Made

- Introduced independent Defect IDs: FR05-BUG-001 through FR05-BUG-005, mapped respectively to FR05-TC-004, FR05-TC-005, FR05-TC-011, FR05-TC-012, and FR05-TC-014.
- Preserved FR05-TC-006 as corrected AUTOMATION_DEFECT history without assigning a PRODUCT_DEFECT ID.
- Standardized titles, report/draft structures, filenames, screenshots, hashtags, and Suggested GitHub Labels according to the approved HW04 Bug Report Standard.
- Normalized descriptive Bug Report content to Vietnamese while retaining English standardized headings, technical identifiers, hashtags, labels, statuses, run IDs, and evidence paths.
- Preserved cross-browser traceability, NOT_PUBLISHED GitHub status, and all approved execution evidence.

Change Illustration

Original style used FR05-TC-005 as the Defect ID and an English description. The corrected bundle uses FR05-BUG-002, identifies FR05-TC-005 under Found by Test Case, uses the standardized title prefix with Vietnamese description, and preserves the same price evidence and requirement semantics.

Correction Notes

The original AI-generated A-009 artifact required human-directed documentation corrections. Its original verdict remains INCOMPLETE; the corrected final bundle was reviewed and approved.

Human Decision Evidence

APPROVE FR05 FINAL DEFECT BUNDLE AFTER FORMAT AND LANGUAGE NORMALIZATION

Artifact A-010 - FR-09 Test Design Bundle

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-09T21:11:31.2308622Z
Workflow Stage	FR-09 compressed requirement analysis and test-case design before human review
Feature / Task	FR-09 - Discount coupons
Related Artifact	docs/test-cases/fr-09/

Verbatim Prompt

[interactions/A-010-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-010-output.md

Verbatim AI Output or Labelled Excerpt

6. Total Test Cases

16.

13. Automation Candidate Count

16.

22. Current Checkpoint

CHECKPOINT: FR09_TEST_DESIGN_BUNDLE_REVIEW_REQUIRED

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	INCOMPLETE
Verdict Scope	Original FR-09 Test Design Bundle before human-directed correction of usage-limit test isolation.

(4) Reasoning

Review Notes

The original AI output correctly extracted the FR-09 five-condition model, created 16 atomic conditions and 16 frontend-oriented test cases, preserved the authoritative Expected Results, and kept implementation discrepancies separate from requirements.

Human review identified one failure-attribution ambiguity: FR09-TC-014 and FR09-TC-015 used total 300000 while the implementation uses total > min_order_amount instead of the authoritative total >= min_order_amount. At the exact minimum, C3 could reject before the usage-limit condition C5 was meaningfully isolated. The original output also listed setup/reset protocol as an open gap; human review approved constrained isolated DB/transaction and controlled frontend-cart setup strategies for the relevant cases. The original verdict therefore remains INCOMPLETE because the pre-correction output required these changes.

(5) Student Fix

Field	Value
Student Decision	MODIFIED
Changes Made	See Changes Made below.
Change Illustration	Usage-limit cases changed from exact-minimum total 300000 to existing verified seed total 4000000 to isolate C5.
Correction Notes	The original AI output required human-directed correction; its Verdict remains INCOMPLETE.
Verification Method	STATIC_DESIGN_REVIEW
Verification Result	PASSED
Final File	docs/test-cases/fr-09/
Approval Status	APPROVED

Changes Made

- Updated FR09-TC-014 to use SAVE10, usage 1/1, existing Keychron seed total 4000000, with C5 false and unchanged payable total 4000000 on rejection.
- Updated FR09-TC-015 to use VIP100, usage 1/2, existing Keychron seed total 4000000, with discount_amount = 100000 and final_amount = 3900000.
- Recorded the APPROVED_WITH_CONSTRAINTS isolated test DB/transaction fixture strategy for FR09-TC-010, FR09-TC-014 and FR09-TC-015.
- Recorded the APPROVED_WITH_CONSTRAINTS controlled frontend cart fixture strategy for approved requirement-supported boundary totals, while preferring existing seed data for normal scenarios.
- Preserved all seven unresolved requirement gaps: exact UI copy, expiry timezone, fractional rounding, usage increment timing, coupon normalization, stacking/removal/replacement and network recovery.
- Revalidated 16 test conditions, 16 test cases, 6 POSITIVE, 5 NEGATIVE, 5 EDGE, 16 automation candidates, 12 FULLY_COVERED atomic requirements and 3 stateful cases.
- Confirmed no new product/database record, test-order dependency, SUT execution or browser execution was introduced.

Human Decision Evidence

APPROVE FR09 TEST DESIGN BUNDLE AFTER C5 ISOLATION CORRECTION

Artifact A-011 - FR-09 Automation Build Bundle

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T04:55:39.0839947+07:00
Workflow Stage	FR-09 compressed automation implementation before execution readiness
Feature / Task	FR-09 - Discount coupons
Related Artifact	tests/fr-09/ test-data/fr-09.json docs/automation-plans/fr-09-automation-plan.md docs/automation-reviews/fr-09-ai-review.md docs/gaps/fr-09-automation-gaps.md

Verbatim Prompt

<interactions/A-011-prompt.md>

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-011-output.md

Verbatim AI Output or Labelled Excerpt

Approved Test Case Count: 16
Implemented Playwright Test Count: 16
Browser Execution Performed: NO
Current Checkpoint: CHECKPOINT: FR09_AUTOMATION_BUILD_REVIEW_REQUIRED

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	INCOMPLETE
Verdict Scope	Original FR-09 Automation Build Bundle before human-directed traceability, negative-oracle and runtime-stability corrections.

(4) Reasoning

Review Notes

The original build correctly retained 16 approved IDs, external datasets, isolated-DB safeguards and failure-evidence configuration. Human review found three material limitations: TC-014/TC-015 could let the known C3 > discrepancy mask C5 usage boundaries; generic rejection depended on an unspecified error-region presentation; and TC-013 evaluated access immediately after navigation. These issues matter because they can create ambiguous failure attribution or false automation failures. The human-review instructions supplied for this artifact define the corrected requirement-preserving boundaries and static verification evidence.

(5) Student Fix

Field	Value
Student Decision	MODIFIED
Changes Made	TC-014/TC-015 retain total 4000000 as a C3-positive control while preserving C5 max/max-1 boundaries; generic rejection now asserts only coupon amounts absent and payable unchanged; TC-013 now waits for the EShop shell and main before mechanism-neutral access evaluation.
Change Illustration	Before: C5 could be masked by the known C3 implementation discrepancy, negative outcomes depended on error CSS, and direct-route visibility was evaluated immediately. After: C5 isolation uses seed total 4000000, error presentation is optional evidence, and TC-013 uses a deterministic render boundary.
Correction Notes	The original A-011 output was materially useful but required human-directed changes before execution readiness. The original Verdict remains INCOMPLETE; the corrected automation build is approved.
Verification Method	STATIC_REVIEW
Verification Result	PASSED
Final File	tests/fr-09/ test-data/fr-09.json docs/automation-plans/fr-09-automation-plan.md docs/automation-reviews/fr-09-ai-review.md docs/gaps/fr-09-automation-gaps.md docs/test-cases/fr-09/
Approval Status	APPROVED

Human Decision Evidence

APPROVE FR09 AUTOMATION BUILD AFTER HUMAN CORRECTIONS

Artifact A-012 - FR-09 Execution Readiness**(1) Prompt + Tool**

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T05:21:00.6489333+07:00
Workflow Stage	FR-09 runtime/isolation verification before multi-browser execution
Feature / Task	FR-09 - Discount coupons - Execution Readiness
Related Artifact	docs/execution-readiness/fr-09-execution-readiness.md

Verbatim Prompt

[interactions/A-012-prompt.md](#)**(2) AI Output**

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-012-output.md

Verbatim AI Output or Labelled Excerpt

Status: READY_FOR_EXECUTION
 Playwright collection: 48 definitions
 Browser execution performed: NO
 Current Checkpoint: CHECKPOINT: FR09_EXECUTION_READINESS_REVIEW_REQUIRED

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-09 Execution Readiness including isolated backend/database preparation, authentication readiness, seed verification, Playwright collection and failure-evidence configuration.

(4) Reasoning

Review Notes

Human review confirmed the isolated backend/database procedure, workspace-DB protection, runtime-only authentication readiness, coupon/product seed oracles, stateful-fixture safety and 48 project-test collection. Evidence configuration preserves only-on-failure screenshots, retain-on-failure traces, video off, zero retries and one FR-09 worker. TC-012 and TC-013 remain NEEDS_MORE_EVIDENCE; no browser execution or PRODUCT_DEFECT classification is claimed. The review instructions supplied for A-012 are the evaluation source.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Changes Made	No human-directed correction was required.
Change Illustration	N/A - execution-readiness artifact was accepted as generated.
Correction Notes	The readiness output is valid for its scope only. Browser execution, failure classification and runtime evaluation of TC-012/TC-013 remain subsequent phases.
Verification Method	STATIC_AND_RUNTIME_READINESS_REVIEW
Verification Result	PASSED
Final File	docs/execution-readiness/fr-09-execution-readiness.md
Approval Status	APPROVED

Human Decision Evidence

APPROVE FR09 EXECUTION READINESS

Artifact A-013 - FR-09 Multi-Browser Execution and Failure Triage**(1) Prompt + Tool**

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T05:44:21.2752065+07:00
Workflow Stage	FR-09 adaptive Chromium/Firefox/WebKit execution before final defect review
Feature / Task	FR-09 - Multi-Browser Execution and Failure Triage
Related Artifact	docs/execution-results/fr-09-chromium-execution.md; docs/execution-results/fr-09-firefox-execution.md; docs/execution-results/fr-09-webkit-execution.md; docs/execution-results/fr-09-cross-browser-summary.md; docs/defects/fr-09/candidate-screenshots/

Verbatim Prompt

[interactions/A-013-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-013-output.md

Verbatim AI Output or Labelled Excerpt

Execution Status: COMPLETED_WITH_PRODUCT_DEFECT_CANDIDATES
 Chromium: 16 total, 10 passed, 6 failed
 Firefox: 16 total, 10 passed, 6 failed
 WebKit: 16 total, 10 passed, 6 failed
 Current Checkpoint: CHECKPOINT: FR09_CROSS_BROWSER_EXECUTION_REVIEW_REQUIRED

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-09 Chromium, Firefox and WebKit execution, runtime failure triage, failure-evidence preservation, database-isolation verification and cross-browser comparison.

(4) Reasoning

Review Notes

Human review xác nhận Chromium, Firefox và WebKit đều hoàn tất 16 / 10 / 6 / 0 (Total / Passed / Failed / Skipped). Sáu Test Case IDs FR09-TC-002, FR09-TC-003, FR09-TC-006, FR09-TC-012, FR09-TC-013 và FR09-TC-016 tái hiện trên cả ba engines, được chấp nhận là PRODUCT_DEFECT_CANDIDATE với confidence HIGH. Không còn AUTOMATION_DEFECT, ENVIRONMENT_FAILURE hoặc NEEDS_MORE_EVIDENCE.

Runtime review đã giải quyết trạng thái NEEDS_MORE_EVIDENCE trước đây của FR09-TC-012 và FR09-TC-013: context chưa xác thực không inject JWT nhưng vẫn tạo cart, tới Checkout và dùng coupon được; shell/main render gate ổn định trong khi coupon controls vẫn usable. Review cũng xác nhận 18 original failure screenshots, 18 traces, sáu Chromium candidate screenshots copy byte-for-byte không recapture, rendered HTML identity cho ba browser, cleanup coupon_usage = 0, isolated backend khỏe và workspace DB SHA-256 không đổi. Human review instructions là evaluation source cho verdict này. Chưa gán final FR09-BUG-xxx.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Changes Made	Không có human-directed correction cho A-013 execution result hoặc failure classifications ban đầu.
Change Illustration	N/A - original multi-browser execution record được chấp nhận as-is.
Correction Notes	Sáu failure vẫn là execution-level PRODUCT_DEFECT_CANDIDATE; verdict này không tạo final Bug ID.
Verification Method	MULTI_BROWSER_RUNTIME_REVIEW
Verification Result	PASSED
Final File	docs/execution-results/fr-09-chromium-execution.md; docs/execution-results/fr-09-firefox-execution.md; docs/execution-results/fr-09-webkit-execution.md; docs/execution-results/fr-09-cross-browser-summary.md; docs/defects/fr-09/candidate-screenshots/
Approval Status	APPROVED

Human Decision Evidence

APPROVE FR09 CROSS BROWSER EXECUTION

Artifact A-014 - FR-09 Final Defect Reporting and Feature Completion Bundle

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T06:14:56.1126884+07:00
Workflow Stage	FR-09 confirmed defect documentation before GitHub Issue publication and feature closure
Feature / Task	FR-09 - Final Defect Reporting and Feature Completion Bundle
Related Artifact	docs/defects/fr-09/; docs/execution-results/fr-09-final-summary.md

Verbatim Prompt

[interactions/A-014-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-014-output.md

Verbatim AI Output or Labelled Excerpt

Confirmed Failed Test Cases: 6
 Confirmed Underlying Defect Count: 5
 Browser Execution Performed: NO
 GitHub Publication Status: NOT_PUBLISHED
 Current Checkpoint: CHECKPOINT: FR09_FINAL_REVIEW_REQUIRED

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-09 final defect reporting, defect deduplication, screenshot promotion, GitHub Issue draft preparation and final feature traceability summary.

(4) Reasoning

Review Notes

Human review xác nhận 16 approved Test Cases và 16 implemented Playwright tests, không có approved test bị blocked. Chromium, Firefox và WebKit đều hoàn tất 16 tests với 10 passed, 6 failed, 0 skipped; cùng sáu Test Case IDs FR09-TC-002, FR09-TC-003, FR09-TC-006, FR09-TC-012, FR09-TC-013, FR09-TC-016 fail trên cả ba engines.

Human review chấp nhận deduplication thành năm product defects: FR09-BUG-001 từ TC-002/TC-003, FR09-BUG-002 từ TC-006, FR09-BUG-003 từ TC-012, FR09-BUG-004 từ TC-013 và FR09-BUG-005 từ TC-016. Năm standardized Bug Reports, năm unpublished GitHub Issue drafts và năm promoted screenshots được xác nhận. Screenshots promoted là byte-identical copies của Chromium failure evidence; sáu candidates, 18 original failure screenshots, 18 traces đều còn nguyên và không có ảnh nào được recapture.

TC-012 và TC-013 là RUNTIME_VERIFIED_MULTI_BROWSER; không còn AUTOMATION_DEFECT, NEEDS_MORE_EVIDENCE hoặc ENVIRONMENT_FAILURE. PASS_WITH_OPEN_REQUIREMENT_GAPS vẫn đúng vì đây documented requirement gaps là documentation boundaries, không invalidates approved FR-09 automation scope. Human review instructions supplied cho A-014 là evaluation source.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Changes Made	Không có human-directed correction nào được yêu cầu.
Change Illustration	N/A - original final defect reporting and feature completion bundle được accepted as-is.
Correction Notes	Human review chấp nhận defect mapping, evidence-promotion policy, unpublished GitHub Issue drafts và final traceability summary.
Verification Method	FINAL_ARTIFACT_REVIEW
Verification Result	PASSED
Final File	docs/defects/fr-09/; docs/execution-results/fr-09-final-summary.md
Approval Status	APPROVED

Verification Notes

- 16 approved Test Cases, 16 scripts, 0 blocked cases.
- Chromium, Firefox và WebKit đều 16 total / 10 passed / 6 failed.
- Có năm Bug Reports, năm GitHub Issue drafts NOT_PUBLISHED, năm promoted screenshots, sáu candidate screenshots, 18 original screenshots và 18 traces.
- Không có browser execution trong finalization; tests, approved test data, Playwright config, SUT source và database không bị finalization sửa.
- Workspace DB protection và final traceability result được human review xác nhận.

Human Decision Evidence

APPROVE FR09 FINAL DEFECT BUNDLE

Artifact A-015 - FR-17 Test Design Bundle and Initial Demo Candidate Analysis**(1) Prompt + Tool**

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T06:47:14.3334053+07:00
Workflow Stage	FR-17 compressed requirement/test design before human review
Feature / Task	FR-17 - Coupon management
Related Artifact	docs/test-cases/fr-17/; docs/demo/fr-17-demo-plan.md

Verbatim Prompt

[interactions/A-015-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-015-output.md

Verbatim AI Output or Labelled Excerpt

```
8. **Total Test Cases:** 16 unique IDs (`FR17-TC-001` - `FR17-TC-016`).
14. **Automation Candidate Count:** 16.
18. **Requirement Coverage Result:** `PASS_WITH_OPEN_REQUIREMENT_GAPS`; 16 `FULLY_COVERED`, 0 `PARTIALLY_COVERED`, 0 `NOT_COVERED`.
24. **Current Checkpoint:** `CHECKPOINT: FR17_TEST_DESIGN_BUNDLE_REVIEW_REQUIRED`.
```

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	INCOMPLETE
Verdict Scope	FR-17 Test Design Bundle and Initial Demo Candidate Analysis.

(4) Reasoning

Review Notes

Human review xác nhận original design có một functional-partition coverage weakness: FR17-TC-003 chỉ chứng minh type selector expose/select percent và fixed, trong khi only fixed was submitted through Coupon Management CREATE. Vì vậy SUT có thể allow selecting percent nhưng reject its CREATE flow mà suite không phát hiện.

Human-directed correction đổi TC-003 thành isolated valid percent CREATE Web Admin UI case, dùng external test-owned data, primary assertions trên created list row/type/value, rồi cleanup ngoài primary objective. TC-004 giữ valid fixed CREATE với approved lower valid numeric controls. Requirement traceability FR17-R02/FR17-R06, state isolation và demo mapping được update; không invent third type, percent maximum hoặc UPDATE behavior. Human review supplied là evaluation source; no SUT/browser/Playwright execution hoặc database mutation was performed.

(5) Student Fix

Field	Value
Student Decision	MODIFIED
Changes Made	TC-003 became isolated valid percent CREATE; TC-004 preserved valid fixed CREATE; FR17-R02/FR17-R06 traceability, state inventory and demo mapping were updated without increasing the test count.
Change Illustration	Original TC-003 inspected/selects the type control only; corrected TC-003 submits a valid owned percent coupon and verifies its observable list state.
Correction Notes	Both authoritative valid type partitions are now exercised through actual CREATE behavior. No arbitrary third type, percent upper boundary or UPDATE behavior was introduced.
Verification Method	STATIC_REVIEW
Verification Result	PASSED
Final File	docs/test-cases/fr-17/; docs/demo/fr-17-demo-plan.md
Approval Status	APPROVED

Verification Notes

- 16 unique Test Conditions; 16 unique Test Cases.
- POSITIVE: 5; NEGATIVE: 8; EDGE: 3.
- Automation Candidate Count: 16 (AUTOMATION_SUITABLE: 1; AUTOMATION_POSSIBLE_WITH_SETUP: 15).
- Stateful Test Count: 12; Read-Only Test Count: 4.
- FR17-TC-003 now verifies valid percent CREATE through Web Admin UI with owned external data and observable Coupon Management list state.
- FR17-TC-004 remains valid fixed CREATE using `discount_value = 1, min_order_amount = 0, max_uses_per_user = 1`.
- FR17-R02 and FR17-R06 are FULLY_COVERED; UPDATE remains NOT_IN_DETAILED_REQUIREMENT_SCOPE; coverage result is PASS_WITH_OPEN_REQUIREMENT_GAPS.
- Demo candidates: TC-004 PRIMARY_DEMO_CANDIDATE; TC-003 and TC-014 SECONDARY_DEMO_CANDIDATE. Final selection remains PENDING until runtime execution evidence exists.
- No SUT, browser or Playwright execution and no database mutation occurred.

Human Decision Evidence

APPROVE FR17 TEST DESIGN AFTER TC003 PERCENT CREATE CORRECTION

-

Artifact A-016 - FR-17 Automation Build Bundle and Demo Automation Preparation

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	Exit code: 0

Wall time: 0.2 seconds Output: 2026-08-10T07:45:10.7122705+07:00 |

Workflow Stage	FR-17 Playwright automation generation before execution readiness and human review
Feature / Task	FR-17 - Coupon management
Related Artifact	test-data/fr-17.json; tests/fr-17/; docs/automation-plans/fr-17-automation-plan.md; docs/automation-reviews/fr-17-ai-review.md; docs/gaps/fr-17-automation-gaps.md; docs/demo/fr-17-demo-plan.md

Verbatim Prompt

[interactions/A-016-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-016-output.md

Verbatim AI Output or Labelled Excerpt

Build Status: BUILT_PENDING_HUMAN_REVIEW; Implemented Playwright Tests: 16; Current Checkpoint: CHECKPOINT: FR17_AUTOMATION_BUILD_REVIEW_REQUIRED.

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	INCOMPLETE
Verdict Scope	FR-17 Playwright Automation Build Bundle and Demo Automation Preparation.

(4) Reasoning

Review Notes

Human review found three AI-generated automation issues in the original FR-17 bundle. First, `getCouponRow()` composed `filter({ has })` through the ancestor coupon table rather than relative to each candidate row. This could miss valid rows and false-fail list/create/delete assertions, including FR17-TC-004 @demo. Second, FR17-TC-005 used an absent-controlled-code rejection path incompatible with the approved duplicate SAVE10 oracle. Third, TC-002 accepted any associated text containing `*` without proving that the indicator was visible.

Human-directed corrections use a row-relative exact-code filter; a duplicate-specific oracle that keeps SAVE10 at exactly one and preserves the total count; and a visible associated label/visible `aria-labelledby` indicator assertion that excludes plain `aria-label` and hidden text. The isolated baseline guard now requires exactly the four approved seed records, and the demo command is PowerShell-compatible. Human review supplied the evaluation source. No browser/SUT runtime execution or database mutation occurred.

(5) Student Fix

Field	Value
Student Decision	MODIFIED
Changes Made	Corrected the relative row locator, TC-005 duplicate-code oracle, TC-002 visible-indicator assertion, exact four-record baseline guard, and Windows demo draft command.
Change Illustration	Original getCouponRow() scoped the inner has locator through the coupon table; corrected implementation evaluates exact code text relative to each coupon row.
Correction Notes	FR17-TC-003 remains valid percent CREATE; FR17-TC-004 remains valid fixed CREATE, PRIMARY_DEMO_CANDIDATE, and the sole @demo test.
Verification Method	STATIC_REVIEW
Verification Result	PASSED
Final File	tests/fr-17/; docs/automation-plans/fr-17-automation-plan.md; docs/automation-reviews/fr-17-ai-review.md; docs/gaps/fr-17-automation-gaps.md; docs/demo/fr-17-demo-plan.md
Approval Status	APPROVED

Verification Notes

- Approved Test Cases: 16; implemented Playwright tests: 16; no missing or duplicate Test Case ID.
- External data remains in test-data/fr-17.json; stateful automation count is 12 and read-only automation count is 4.
- TC-005 now asserts total coupon count remains at baseline and SAVE10 remains exactly once. TC-002 requires an actual visible required-field indicator.
- Workspace DB guard and deterministic, test-owned stateful cleanup remain enabled.
- Screenshot: only-on-failure; trace: retain-on-failure; video: off; retries: 0; future FR-17 workers: 1.
- Primary DEMO_FIX_CANDIDATE: FR-17 getCouponRow relative-locator correction; Type: TEST_SCRIPT_CORRECTION; Runtime Verification: PENDING.
- No browser execution, SUT runtime execution, or database mutation occurred.

Human Decision Evidence

APPROVE FR17 AUTOMATION BUILD AFTER HUMAN-DIRECTED SCRIPT CORRECTIONS

-

Artifact A-017 - FR-17 Execution Readiness and Demo Runtime Preparation**(1) Prompt + Tool**

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T08:30:35.2002362+07:00
Workflow Stage	FR-17 isolated runtime and Playwright collection verification before multi-browser execution
Feature / Task	FR-17 - Coupon management
Related Artifact	docs/execution-readiness/fr-17-execution-readiness.md; tests/fr-17/; test-data/fr-17.json; docs/demo/fr-17-demo-plan.md

Verbatim Prompt

<interactions/A-017-prompt.md>

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-017-output.md

Verbatim AI Output or Labelled Excerpt

Readiness Status: READY_FOR_EXECUTION; Total Project-Test Definitions: 48; Current Checkpoint: CHECKPOINT: FR17_EXECUTION_READINESS_REVIEW_REQUIRED.

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-17 Execution Readiness and Demo Runtime Preparation.

(4) Reasoning

Review Notes

Human review xác nhận readiness READY_FOR_EXECUTION: Web Admin tại <http://localhost:5174> và isolated backend tại <http://localhost:3000> đều healthy. Backend sử dụng isolated directory/database của run fr17-readiness-20260810T082458317+0700; workspace backend/database.sqlite bị cấm cho fixtures và SHA-256 vẫn là 2251BA251C2B0722F0E524775C6DB5FC1B7CFB046D561BA26905B1EDCC52BA02.

Baseline isolated chính xác gồm bốn coupons SAVE10, BIGBUY, VIP100, EXPIRED; required schema, admin/non-admin authentication và không lưu credentials/password/JWT đều đã được review. Playwright collection xác nhận 16 tests mỗi browser (48 project-test definitions), không missing/duplicate IDs. FR17-TC-004 là @demo duy nhất với 3 project-test definitions; demo command collection pass. FR-17 getCouponRow relative-locator correction được static-wired, còn runtime verification PENDING đúng phạm vi readiness.

Human review cũng xác nhận workers = 1, retries = 0, screenshot only-on-failure, trace retain-on-failure, video off và HTML report identity readiness pass. Không có FR-17 browser/business-test execution hay database business mutation. Human review instructions là evaluation source cho verdict này.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Changes Made	Không có human-directed correction được yêu cầu.
Change Illustration	N/A - original readiness record được chấp nhận as-is.
Correction Notes	DEMO_FIX runtime verification vẫn PENDING cho tới execution.
Verification Method	STATIC_AND_RUNTIME_READINESS_REVIEW
Verification Result	PASSED
Final File	docs/execution-readiness/fr-17-execution-readiness.md
Approval Status	APPROVED

Human Decision Evidence

APPROVE FR17 EXECUTION READINESS

Artifact A-018 - FR-17 Cross-Browser Execution, Failure Triage and Demo Fix Runtime Verification

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T08:51:36.9311488+07:00
Workflow Stage	FR-17 approved automation execution on Chromium, Firefox and WebKit before final defect documentation
Feature / Task	FR-17 - Coupon management
Related Artifact	docs/execution-results/fr-17-chromium-execution.md; docs/execution-results/fr-17-cross-browser-summary.md; docs/defects/fr-17/candidate-screenshots/; docs/demo/fr-17-demo-plan.md

Verbatim Prompt

<interactions/A-018-prompt.md>

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-018-output.md

Verbatim AI Output or Labelled Excerpt

Execution Status: STOPPED_AT_CHROMIUM_AUTOMATION_DEFECT_GATE; Chromium: 16 / 13 / 3 / 0; Automation Defect: FR17-TC-003; Current Checkpoint: CHECKPOINT: FR17_CHROMIUM_AUTOMATION_REVIEW_REQUIRED.

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-17 Chromium execution, runtime failure triage, evidence preservation, database-safety verification and adaptive execution gate decision.

(4) Reasoning

Review Notes

Human review xác nhận Chromium hoàn tất 16 / 13 / 3 / 0; accepted failure classifications là FR17-TC-002 và FR17-TC-016 PRODUCT_DEFECT_CANDIDATE, cùng FR17-TC-003 AUTOMATION_DEFECT. TC-002 đã reach Coupon Management rồi mới thiếu visible required-field indicator. TC-003 tạo thành công valid percent coupon, nhưng AI-generated assertion bổ sung locale-dependent cho min-order nằm ngoài primary approved verification scope nên tạo false failure. TC-016 đã runtime-verify valid non-admin identity và authorization boundary fail không có automation/environment setup failure.

Chromium automation-defect gate vì vậy chặn Firefox/WebKit đúng quy định. Human review cũng xác nhận không có automatic correction/rerun, hai candidate screenshots, ba original screenshots, ba traces, PASS_RENDERED HTML report, isolated DB cleanup/baseline và workspace DB protection. TC-004 @demo PASS trên Chromium và corrected getCouponRow path được exercised; DEMO_FIX chỉ PARTIALLY_VERIFIED vì cross-browser execution bị chặn. Human review instructions là evaluation source cho verdict này.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Changes Made	Không có correction cho original A-018 execution result hoặc failure classification.
Change Illustration	N/A - original Chromium result, triage và adaptive gate decision được chấp nhận as-is.
Correction Notes	Multi-browser verification vẫn pending vì Firefox/WebKit bị chặn đúng bởi AUTOMATION_DEFECT gate.
Verification Method	RUNTIME_EXECUTION_REVIEW
Verification Result	PASSED
Final File	docs/execution-results/fr-17-chromium-execution.md; docs/execution-results/fr-17-cross-browser-summary.md; docs/defects/fr-17/candidate-screenshots/; docs/demo/fr-17-demo-plan.md
Approval Status	APPROVED

Verification Notes

- Candidate product screenshots preserved: 2; original Chromium failure screenshots: 3; trace evidence: 3.
- Chromium HTML report: PASS_RENDERED; isolated DB cleanup: PASS; final baseline: SAVE10, BIGBUY, VIP100, EXPIRED.
- Workspace DB SHA-256 remains 2251BA251C2B0722F0E524775C6DB5FC1B7CFB046D561BA26905B1EDCC52BA02 (PASS).
- FR17-TC-004 @demo PASSes on Chromium and DEMO_FIX_RUNTIME_VERIFIED_CHROMIUM is YES; DEMO_FIX Runtime Verification is PARTIALLY_VERIFIED.
- The earlier npx.cmd EINVAL launcher diagnostic started no browser and is excluded from business execution metrics.

Human Decision Evidence

APPROVE FR17 CHROMIUM EXECUTION AND AUTOMATION DEFECT GATE

Artifact A-019 - FR-17 TC-003 Runtime Automation Defect Correction**(1) Prompt + Tool**

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T09:07:57.7939103+07:00
Workflow Stage	Human-directed correction after Chromium automation-defect gate and before corrected multi-browser execution
Feature / Task	FR-17 - Coupon management
Related Artifact	tests/fr-17/; docs/automation-reviews/fr-17-ai-review.md; docs/demo/fr-17-demo-plan.md; docs/execution-results/fr-17-chromium-execution.md

Verbatim Prompt

[interactions/A-019-prompt.md](#)

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-019-output.md

Verbatim AI Output or Labelled Excerpt

Correction Status: CORRECTED_PENDING_HUMAN_REVIEW; Affected Test Case: FR17-TC-003; Browser Execution: NO;
Current Checkpoint: CHECKPOINT: FR17_TC003_AUTOMATION_CORRECTION_REVIEW_REQUIRED.

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-17 TC-003 runtime automation-defect correction after the Chromium automation-defect gate.

(4) Reasoning

Review Notes

Human review xác nhận correction giữ nguyên approved TC-003 business objective: row owned tồn tại đúng một lần, visible, đúng percent type và submitted discount_value. Generic helper cũ đã assert min-order display qua Number(...).toLocaleString() sau khi create thành công, dù assertion này không thuộc primary Expected Result; correction loại bỏ false-failure oracle đó.

TC-004 vẫn là PRIMARY_DEMO_CANDIDATE và sole @demo, giữ các lower-bound checks discount_value=1, min_order_amount=0, max_uses_per_user=1. Primary getCouponRow relative-locator fix vẫn PARTIALLY_VERIFIED trên Chromium; TC-003 assertion-scope/locale-independence là secondary correction riêng, runtime PENDING. Human review xác nhận 16 unique Playwright tests, A-018 evidence manifest/workspace DB hash không đổi, và phase không chạy browser hay business DB mutation. Human review instructions là evaluation source cho verdict này.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Changes Made	Không có additional human-directed correction cho A-019 correction output.
Change Illustration	N/A - TC-003 core/boundary split được chấp nhận as-is.
Correction Notes	Runtime verification của secondary correction vẫn pending cho tới approved corrected execution.
Verification Method	STATIC_REVIEW
Verification Result	PASSED
Final File	tests/fr-17/helpers/coupon-ui.js; tests/fr-17/fr-17.spec.js; docs/automation-reviews/fr-17-ai-review.md; docs/automation-plans/fr-17-automation-plan.md; docs/gaps/fr-17-automation-gaps.md; docs/demo/fr-17-demo-plan.md
Approval Status	APPROVED

Verification Notes

- FR17-TC-003 no longer uses min-order, max-use or expiry display as PASS/FAIL oracles; no affected row assertion uses Number(...).toLocaleString().
- A-018 aggregate evidence manifest remains
4A7CBF716A12EBC41F2C888A016988FE13366199346895F2F44B4A1FEE60DAF3; workspace DB SHA-256 remains
2251BA251C2B0722F0E524775C6DB5FC1B7CFB046D561BA26905B1EDCC52BA02.
- No browser execution and no database business mutation occurred during correction.

Human Decision Evidence

APPROVE FR17 TC003 AUTOMATION CORRECTION

Artifact A-020 - FR-17 Corrected Cross-Browser Execution and Demo Runtime Verification

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T09:31:21.0679012+07:00
Workflow Stage	FR-17 corrected Chromium execution followed by Firefox/WebKit after approved TC-003 automation correction
Feature / Task	FR-17 - Coupon management
Related Artifact	docs/execution-results/fr-17-chromium-execution.md; docs/execution-results/fr-17-chromium-corrected-execution.md; docs/execution-results/fr-17-firefox-execution.md; docs/execution-results/fr-17-webkit-execution.md; docs/execution-results/fr-17-cross-browser-summary.md; docs/defects/fr-17/candidate-screenshots/; docs/demo/fr-17-demo-plan.md

Verbatim Prompt

<interactions/A-020-prompt.md>

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-020-output.md

Verbatim AI Output or Labelled Excerpt

Execution Status: COMPLETED_CORRECTED_MULTI_BROWSER_EXECUTION_WITH_PRODUCT_DEFECT_CANDIDATES; final failure-set comparison: PARTIALLY_OVERLAPPING; Current Checkpoint: CHECKPOINT: FR17_CORRECTED_CROSS_BROWSER_EXECUTION_REVIEW_REQUIRED.

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-17 corrected Chromium, Firefox and WebKit execution, runtime failure triage, correction verification, evidence preservation, database safety and demo runtime verification.

(4) Reasoning

Review Notes

Human review xác nhận historical Chromium run fr17-chromium-20260810T084402107+0700 được preserve unchanged và historical FR17-TC-003 automation failure không được tính vào final product metrics. Corrected Chromium, Firefox và WebKit đều được review theo actual runtime evidence; không có human-directed correction nào cần thiết cho A-020 execution result, failure classification hoặc cross-browser comparison.

Final failure set là PARTIALLY_OVERLAPPING: FR17-TC-002 và FR17-TC-016 reproduce trên cả ba engines; FR17-TC-008 và FR17-TC-011 chỉ reproduce trên corrected Chromium, trong khi Firefox/WebKit pass. Review không suy luận backend root cause độc lập trình duyệt cho TC-008/TC-011 chỉ từ static implementation observations.

FR17-TC-003 assertion-scope / locale-independence correction và primary getCouponRow relative-locator DEMO_FIX đều VERIFIED_MULTI_BROWSER. FR17-TC-004 @demo pass trên Chromium, Firefox và WebKit, nên FR17-TC-004 là Primary Demo Test với Demo Runtime Status VERIFIED_MULTI_BROWSER.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Changes Made	Không có human-directed correction nào cần thiết cho A-020 execution result, failure classifications hoặc cross-browser comparison.
Change Illustration	N/A - corrected multi-browser execution result được chấp nhận as-is.
Correction Notes	Historical A-018 evidence, original A-020 prompt/output và final corrected execution evidence được preserve.
Verification Method	CORRECTED_MULTI_BROWSER_RUNTIME_REVIEW
Verification Result	PASSED
Final File	docs/execution-results/fr-17-chromium-corrected-execution.md; docs/execution-results/fr-17-firefox-execution.md; docs/execution-results/fr-17-webkit-execution.md; docs/execution-results/fr-17-cross-browser-summary.md; docs/defects/fr-17/candidate-screenshots/; docs/demo/fr-17-demo-plan.md
Approval Status	APPROVED

Verification Notes

- Corrected Chromium fr17-chromium-corrected-20260810T092053847+0700: 16 total / 12 passed / 4 failed / 0 skipped; accepted product candidates are FR17-TC-002, FR17-TC-008, FR17-TC-011, FR17-TC-016.
- Firefox fr17-firefox-20260810T092258944+0700 và WebKit fr17-webkit-20260810T092419436+0700: mỗi run 16 total / 14 passed / 2 failed / 0 skipped; accepted product candidates là FR17-TC-002, FR17-TC-016.
- Remaining AUTOMATION_DEFECT, NEEDS_MORE_EVIDENCE và ENVIRONMENT_FAILURE: None.
- FR17-TC-003 passes on Chromium, Firefox và WebKit; secondary correction is VERIFIED_MULTI_BROWSER.
- FR17-TC-004 @demo passes on all three engines; primary demo test is FR17-TC-004, Demo Runtime Status và primary DEMO_FIX Runtime Verification đều VERIFIED_MULTI_BROWSER.
- Chromium, Firefox và WebKit HTML reports are PASS_RENDERED, each visibly showing FR-17, Run by: 23127107, ISO timestamp, Run ID và browser identity.
- Isolated DB cleanup is PASS: exact baseline SAVE10, BIGBUY, VIP100, EXPIRED; owned FR17 residue 0. Workspace DB protection is PASS with SHA-256 2251BA251C2B0722F0E524775C6DB5FC1B7CFB046D561BA26905B1EDCC52BA02.
- Original A-018 historical evidence preservation và candidate screenshot preservation are PASS. No test code, Playwright config, SUT source hoặc workspace database was modified during A-020 execution.

Human Decision Evidence

APPROVE FR17 CORRECTED MULTI BROWSER EXECUTION

Artifact A-021 - FR-17 Final Defect Reporting, Feature Completion and Demo Readiness Bundle

(1) Prompt + Tool

Field	Value
Tool	Codex
Model	GPT-5
Date and Time	2026-08-10T09:49:25.7457840+07:00
Workflow Stage	FR-17 confirmed defect documentation and recording preparation before final human review
Feature / Task	FR-17 - Coupon management
Related Artifact	docs/defects/fr-17/; docs/execution-results/fr-17-final-summary.md; docs/demo/fr-17-demo-plan.md; docs/demo/fr-17-demo-checklist.md

Verbatim Prompt

<interactions/A-021-prompt.md>

(2) AI Output

Field	Value
Output Storage	EXTERNAL_FILE
Full Output / Evidence Path	interactions/A-021-output.md

Verbatim AI Output or Labelled Excerpt

Finalization Status: COMPLETED_PENDING_HUMAN_REVIEW; confirmed underlying defect count: 4; recording status: NOT_RECORDED; GitHub publication status: NOT_PUBLISHED; Current Checkpoint: CHECKPOINT: FR17_FINAL_REVIEW_REQUIRED.

(3) Verdict

Field	Value
Review Status	FINALIZED
Verdict	VALID
Verdict Scope	FR-17 final defect reporting, evidence promotion, final feature traceability, demo preparation and recording readiness.

(4) Reasoning

Review Notes

Human review xác nhận A-021 finalization output hợp lệ và không cần human-directed correction. Final corrected execution gồm 48 project-test combinations: Chromium 16 total / 12 passed / 4 failed / 0 skipped, Firefox 16 / 14 / 2 / 0, WebKit 16 / 14 / 2 / 0; tổng cộng 40 passed / 8 failed / 0 skipped.

Bốn confirmed failed Test Case IDs được deduplicate đúng thành bốn product defects: FR17-BUG-001 <- FR17-TC-002, FR17-BUG-002 <- FR17-TC-008, FR17-BUG-003 <- FR17-TC-011, FR17-BUG-004 <- FR17-TC-016. Historical FR17-TC-003 không được đưa vào product-defect mapping vì automation defect ban đầu đã được sửa và verified multi-browser. FR17-BUG-001 và FR17-BUG-004 là CROSS_BROWSER; FR17-BUG-002 và FR17-BUG-003 là CHROMIUM_SPECIFIC_OBSERVED_BEHAVIOR, không suy luận universal backend root cause từ static observations.

Human review cũng xác nhận bốn standardized Bug Reports, bốn GitHub Issue drafts, promoted screenshots byte/hash-identical với approved candidate sources, và original execution evidence được preserve. Remaining AUTOMATION_DEFECT, NEEDS_MORE_EVIDENCE, ENVIRONMENT_FAILURE đều bằng 0. Final traceability là PASS_WITH_OPEN_REQUIREMENT_GAPS; 11 requirement gaps tiếp tục là documentation boundaries, không bị chuyển thành invented requirements. FR17-TC-004 @demo, primary getCouponRow DEMO_FIX, và secondary FR17-TC-003 runtime correction đều VERIFIED_MULTI_BROWSER. Demo Recording Checklist là READY; recording vẫn NOT_RECORDED và GitHub publication vẫn NOT_PUBLISHED.

(5) Student Fix

Field	Value
Student Decision	ACCEPTED
Changes Made	Không có human-directed correction nào cần thiết cho A-021 finalization output.
Change Illustration	N/A - final defect, evidence, traceability và demo-readiness bundle được chấp nhận as-is.
Correction Notes	Original verbatim A-021 prompt/output và execution evidence được preserve unchanged.
Verification Method	FINAL_ARTIFACT_AND_DEMO_READINESS_REVIEW
Verification Result	PASSED
Final File	docs/defects/fr-17/; docs/execution-results/fr-17-final-summary.md; docs/demo/fr-17-demo-plan.md; docs/demo/fr-17-demo-checklist.md
Approval Status	APPROVED

Verification Notes

- Approved Test Cases: 16; Automated Test Cases: 16; Blocked: 0.
- Final project-test metrics: 48 total / 40 passed / 8 failed / 0 skipped.
- Confirmed failed Test Case IDs: FR17-TC-002, FR17-TC-008, FR17-TC-011, FR17-TC-016; confirmed underlying product defects: 4.
- Final mapping: FR17-BUG-001 <- FR17-TC-002, FR17-BUG-002 <- FR17-TC-008, FR17-BUG-003 <- FR17-TC-011, FR17-BUG-004 <- FR17-TC-016.
- Cross-browser result: FR17-BUG-001 and FR17-BUG-004 are CROSS_BROWSER; FR17-BUG-002 and FR17-BUG-003 are CHROMIUM_SPECIFIC_OBSERVED_BEHAVIOR.
- Four Bug Reports and four GitHub Issue drafts exist; GitHub Publication Status: NOT_PUBLISHED.
- Screenshot promotion is PASS; four promoted screenshots are byte/hash-identical to approved candidate sources, and original execution evidence remains preserved.
- Remaining AUTOMATION_DEFECT, NEEDS_MORE_EVIDENCE, and ENVIRONMENT_FAILURE: 0.
- Final Traceability: PASS_WITH_OPEN_REQUIREMENT_GAPS; the existing 11 requirement gaps remain open.
- Primary Demo Test: FR17-TC-004; Demo Tag: @demo; Demo Runtime Status: VERIFIED_MULTI_BROWSER.
- Primary DEMO_FIX (getCouponRow relative-locator correction) and secondary FR17-TC-003 assertion-scope/locale-independence correction are VERIFIED_MULTI_BROWSER.
- Demo Recording Checklist: READY; Recording Status: NOT_RECORDED.
- Feature Status: FR17_COMPLETE; Can Begin Recording: YES.
- No browser execution or database mutation occurred during A-021 finalization; workspace database protection was preserved.

Human Decision Evidence

APPROVE FR17 FINAL DEFECT AND DEMO READINESS BUNDLE
<!-- AUDIT_ENTRIES_END -->

4. Summary of AI Accuracy

<!-- AUDIT_SUMMARY_START -->

Metric	Count	Percentage
Total AI-generated artifacts audited	21	100.00%
VALID - correct, accepted as-is	12	57.14%
INVALID - wrong, rejected	0	0.00%
INCOMPLETE - acceptable after edits	9	42.86%

Calculation rule

Percentage = verdict count / total finalized artifacts × 100
Entries with PENDING_HUMAN_REVIEW are excluded from finalized verdict totals.
<!-- AUDIT_SUMMARY_END -->

5. Conclusion - When Should AI Be Used or Not?

Write 80-150 words after audit entries and human review exist.

Write the conclusion here.

6. Mandatory Disclosure

Replace the bracketed parts before submission:

“[Test cases / script / dataset / report] was initially generated by [AI tool name]; I reviewed and modified [section X], added [edge cases Y, Z]; [section W] was written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category.”

7. Student Confirmation

Field	Value
Student name (printed)	
Student ID	
Class / Cohort	
Course	CS423 / CSC13003 - Software Testing
Instructor	
Date	
Signature	

8. References

- FIT@HCMUS. _AI Audit Report - 5-section Template per Artifact_.
- ISTQB. _Foundation Level Syllabus_, latest applicable version.
- Course slides and technical sources cited in individual audit entries.